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Department of Computer Science and Engineering (CSE-1)

A.Y:2026-27 **IV B.Tech- Odd Semester** **Y23 Batch**

Capstone Project – I Abstract Proforma

Date:

Batch Number: 3 **Project Usecase No:** Cloud 225	**Project Title:** Emerging Infrastructure Project Maturity Assessment for Adoption Decisions	**Guide Name:** K Swarna Mam **Guide emp. Id:**4935

Project Abstract:

Open-source cloud infrastructure projects are widely used by organizations to build and deploy modern applications. However, selecting a reliable project is difficult because common GitHub indicators such as stars, forks, and repository popularity do not necessarily represent its long-term sustainability. Existing studies have examined project maintenance and software security separately, but there is a need for an integrated approach that evaluates the maturity and sustainability of projects for organizational adoption.

This project proposes an Emerging Infrastructure Project Maturity Assessment for Adoption Decisions system that evaluates GitHub repositories using technical and sustainability indicators beyond popularity. The system collects repository data through the GitHub API and analyzes factors such as maintainer distribution, contributor activity, release continuity, issue resolution, development activity, and security practices. These indicators are used to assess the current maturity of a project and to develop a model that predicts its maintenance continuity, helping identify projects that may become inactive or difficult to maintain.

The proposed system combines repository analytics, sustainability assessment, maintenance prediction, and enterprise adoption risk assessment into a single decision-support platform. It generates an Enterprise Readiness Score and Adoption Risk Score and provides an understandable recommendation such as Adopt, Review Carefully, or Avoid. The system is validated using historical projects that later became stalled or unmaintained, with the aim of helping organizations make more informed decisions when selecting open-source cloud infrastructure projects and reducing the risk of adopting projects with poor long-term sustainability.

To be filled by the Guide:

Student IDs:	**Attendance**	**Student's Understanding Level on Project (V. Good: 10-8M Good: 7-6M Average: 5-4M Poor: 3-0M)**
2300030935		
2300033750		
2300033246		
2300031717		

Guide Remarks:

Guide Signature: